Reflex – Architecture Design Document
Week 3 · Readiness Sprint
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Purpose: To explain, in clear terms, how Reflex is designed: what technologies we use, what data we store, how a delivery is assigned, and how status moves from start to finish.

1. Problem

Small Kenyan retailers (electronics shops, pharmacies, hardware stores) currently organise deliveries over WhatsApp and phone calls. There is no clear record of who is assigned, no shared status, and no proof of delivery.

2. Stack – What / How / Why

Layer	What we use	How it is used	Why we chose it
Live System	Deployed on Vercel	Available at https://reflexsuport.vercel.app	Easy to put online, free for a pilot, works on phones so shop staff and riders can use it without installing an app
Frontend	React web app	Screens for retailer staff, dispatcher, and rider	Works in a normal browser on low-end phones; no app store needed
Database	Supabase (Postgres)	Stores users, customers, delivery requests, and status history	Free to start, reliable, and gives one clear place where the truth about every delivery lives
Roles	Users table with role: retailer_staff, dispatcher, or rider Controls who can create requests, who can assign, and who can update status Stops the wrong person doing the wrong action	Controls who can create requests, who can assign, and who can update status	Stops the wrong person doing the wrong action
Visibility	Screens refresh so people can see current status Retailer and	Retailer and dispatcher can check where a delivery stands	Answers the main pain: “Where is my delivery?”

	dispatcher can check where a delivery stands Answers the main pain: “Where is my delivery?”		

Why not stay on WhatsApp only?
WhatsApp has no structured assignment, no forced status steps, and no permanent record. Reflex keeps the same people (shop, dispatcher, rider) but adds a clear system of record.

3. <u>Data Model – What / How / Why</u>

Database: Supabase

Tables

Users
- id
- name
- role (dispatcher / rider / retailer_staff)

Why: One list of everyone in the system and what they are allowed to do.

Customers
- id
- name
- phone
- address

Why: The person receiving the package. Stored separately so details are not retyped every time.

DeliveryRequests
- id
- customer_id (link to Customers)
- item_description
- status (default: open)
- created_by (link to Users – the retailer staff who logged it)
- assigned_rider_id (link to Users – the rider, empty until assigned)
- created_at

Why: This is the main delivery job. It shows what is being delivered, who asked for it, who is taking it, and the current status.

StatusLogs
Records of status changes over time

Why: Gives a history trail (proof of what happened and when). Useful if there is a dispute.

How the tables connect
- A retailer staff (User) creates a DeliveryRequest
- That request is linked to a Customer
- A dispatcher (User) assigns a rider (User)
- Status changes can be recorded in statusLogs

4. Assignment Flow – What / How / Why

This follows the team's Delivery Management Flowchart.

Step by step

1.**Retailer staff** logs a new delivery request.
- Enters customer name, phone, address, item description
- Status becomes **Open**
- No rider is assigned yet

2.**Dispatcher** opens the list of open requests

Sees all requests with status **Open**

3.**Dispatcher** selects a request and chooses a rider
- System sets assigned_rider_id
- Status becomes Assigned

4.**Rider** opens "My deliveries"
- Sees only the deliveries assigned to them

Rules we follow
- Only a dispatcher can assign a rider
- One delivery has only one rider at a time
- Assignment is recorded (who was given the job)

Why this way?

In WhatsApp, anyone can claim a job or nobody knows who took it.
Here, assignment is deliberate and visible.

5. Status Flow – What / How / Why

Allowed sequence

Open — Assigned —Picked Up —Delivered

Status	Meaning
Open	New request created by retailer staffMeaning
Assigned	Dispatcher has given it to a rider
Picked Up	Rider has collected the item
Delivered	Item has been given to the customer

Who can change status

Change	Who can do it
Open — Assigned	Dispatcher only
Assigned — Picked Up	The assigned rider
Picked Up — Delivered	The assigned rider

Why a strict order?
WhatsApp lets anyone type “delivered” with no proof.
A fixed sequence plus a history log gives the shop a trustworthy answer to “Where is my delivery?”

6. End-to-end picture

- Retailer staff creates request — status **Open**
- Dispatcher assigns rider — status **Assigned**
- Rider picks up item — status **Picked Up**
- Rider delivers to customer — status **Delivered**
- Retailer can check status at any time on the system

Live system: https://reflexsuport.vercel.app

7. Design principles (what we are defending)

- **One source of truth** – the database, not chat messages
- **Clear roles** – only the right person can assign or update status
- **Simple enough for real shops** – must be faster and clearer than WhatsApp, not harder
- **Explainable choices** – every part of the design can be justified in plain language for the panel

8. What we are not solving in this version

- Automatic SMS/WhatsApp notifications
- Live GPS tracking of the rider
- Payments or cash-on-delivery collection
- Multiple branches or large-scale fleet management